PAPER_1078: QCalcGeom Master Equation Derivation

Star Magic UQFF Framework — Session 225

Author: Daniel Murphy **Date:** 2026 **Module:** `qcalcgeom_{core\_derivation}.py`

Abstract

We derive the QCalcGeom master equation combining Breit-frame Schwarzschild-flux-graviton (BSFG) crossover radius, 26-dimensional compactification scale, Ramanujan polylogarithmic corrections, and SCm phonon fluence into a single length-scale observable:

$$\text{QCalcGeom}(r, \Gamma) = r_{\text{cross}} \cdot (26!)^{-1/13} \cdot S_{26}^{(3)}([SSq]) \cdot \Phi(\omega, \Gamma)$$

1. BSFG Crossover Radius

The crossover radius marks the transition between quantum-gravitational and classical regimes:

$$r_{\text{cross}} = \sqrt{\eta_{\text{BSFG}}} \cdot \frac{GM}{c^2}$$

where $\eta_{\text{BSFG}} = 10^{-22}$ is the BSFG coupling and GM/c^2 is the gravitational radius. For the Sun ($M = 1.989 \times 10^{30}$ kg):

$$r_{\text{cross}} = \sqrt{10^{-22}} \cdot \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^{8})^2} = 1.477 \times 10^{-8} \text{ m}$$

2. 26D Compactification Scale

The compactification scale arises from the 26-dimensional string compactification volume:

$$\ell_c = (26!)^{-1/13} = (4.033 \times 10^{26})^{-1/13} = 8.983 \times 10^{-3}$$

This sets the dimensionless ratio between compact and extended dimensions. The first five Kaluza-Klein eigenvalues are $\lambda_n = n(n + 25)$ for $n = 1, \dots, 5$: {26, 54, 84, 116, 150}.

3. Ramanujan Polylogarithmic Sum

The $S_{26}^{(3)}$ correction factor:

$$S_{26}^{(3)}([SSq]) = \sum_{n=1}^{26} \frac{[SSq]^n}{n^{26}} \cdot R_n^{(3)}$$

where $R_n^{(3)}$ is the Ramanujan correction factor. With $[SSq] = 0.57$:

$$S_{26}^{(3)} = 9.500 \times 10^{-2}$$

4. Phonon Fluence Factor

$$\Phi(\omega, \Gamma) = \exp\left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_G^2}\right) \cdot S_{26}^{(3)}$$

This Gaussian envelope peaks at $\Gamma_0 = 2\pi \times 0.10$ THz with width $\sigma_G = 0.08 \times 2\pi \times 10^{12}$ rad/s.

5. Master Equation Assembly

Combining all four factors:

$$\text{QCalcGeom}(M_\odot, \Gamma_0) = 1.477 \times 10^{-8} \times 8.983 \times 10^{-3} \times 9.500 \times 10^{-2} \times 9.500 \times 10^{-2}$$

$$= 1.197 \times 10^{-12} \text{ m}$$

6. Dimensional Proof

The SI unit trace confirms correct dimensionality:

- r_{cross}: [m] (length)
- $(26!)^{-1/13}$: dimensionless
- $S_{26}^{(3)}$: dimensionless
- Φ: dimensionless
- Product: [m] – a **length scale**

7. Validation

- Linewidth sweep shows peak at $\Gamma = 0.10$ THz
- Linear mass scaling: QCalcGeom($2M$) / QCalcGeom(M) = 2.0000
- All 10 self-tests pass

References

1. BSFG coupling: `qcalcgeom_helpers.py` ($\eta = 10^{-22}$)
2. 26D string compactification: `MAIN_{1\_CoAnQi}.cpp` SOURCE115-116
3. Ramanujan corrections: `vds_{dvp\_bsh\_symbolic\_proofs}.py`
4. SCm phonon framework: `scm_{phonon\_linewidth}.py`

Session 225: Late-Corpus Physics Integration (PAPER_1000-1081)

> *The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and S26(3) Ramanujan corrections into this paper's domain.*

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\mathrm{SSq}]| < 1$ (satisfied by $[\mathrm{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\mathrm{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\mathrm{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1080	Ramanujan Binomial Expansion Proof R_n^{(26,3)}

2 cross-reference(s) identified.